2025 5th International Conference on Electronic Communication, Computer Science and Technology (ECCST) | 979-8-3315-8002-5/25/$31.00 ©2025 IEEE | DOI: 10.1109/ECCST68196.2025.11441191

Research on Crop Pest Recognition Based on Improved YOLOv8

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***Abstract*—Aiming at the problems of numerous types, complex information of crop pests and low efficiency of traditional manual monitoring methods, this paper proposes an improved crop pest recognition method based on YOLOv8. Three optimization strategies are adopted: modifying the input size, adding the SE (Squeeze-and-Excitation) attention mechanism to the detection head, and adding the ECA (Efficient Channel Attention) attention mechanism to the backbone network. A dataset containing 2,470 images of ten common crop pests (including mole cricket, sugar beet weevil, fall armyworm) is constructed for model training and validation. Experimental results show that the improved model achieves significant performance improvement: the precision of the original YOLOv8 model is 80%, while the precisions of the models optimized by the three strategies reach 82%, 92% and 83.5% respectively. The model integrated with the SE attention mechanism exhibits the best performance, with fast inference speed and strong robustness, providing an effective technical solution for rapid and accurate identification of crop pests.**

Keywords—crop pests, YOLOv8, attention mechanism, target detection, deep learning

I. Introduction

As one of the major agricultural countries in the world, China produces nearly a quarter of the world's grain and solves the food and clothing problem for nearly one-fifth of the world's population[1]. However, crop pests have always been a key factor restricting agricultural production. According to the statistics of the Food and Agriculture Organization of the United Nations (FAO), pests cause losses of about 20% for food crops, 25% for tea plants, 30% for cotton and 40% for fruit trees worldwide every year[1]. In severe cases, local crop yield reduction can even exceed 50%, bringing huge economic losses to farmers.

Traditional crop pest identification mainly relies on manual field investigation, which requires professional entomological knowledge and rich practical experience. This method is not only time-consuming and labor-intensive, but also subjective and prone to misjudgment, making it difficult to meet the needs of large-scale, efficient and intelligent modern agricultural management[2]. With the rapid development of deep learning and computer vision technology, convolutional neural networks (CNN) have been widely applied in crop pest recognition due to their powerful automatic feature extraction capabilities[6].

Domestic and foreign scholars have conducted extensive research in this field. Gui Yupeng et al.[3] proposed a lightweight YOLOv8-Rice algorithm for rice pest detection, which reduced the model size by 61.7% while ensuring an average precision of 94.1%. Fragoso J et al.[4] used YOLO series models for coffee pest detection, and YOLOv8s achieved the best real-time performance with an mAP of 54.5%. Bijlwan A et al.[5] adopted EfficientNetB0 model for rice pest classification, achieving a test accuracy of 99.45%. However, existing studies mostly focus on single or a few types of pests, and the data samples have high quality and few interference factors. In practical application scenarios, there are many types of pests, and the samples cover multiple perspectives and growth stages, which increases the difficulty of feature extraction and classification, leading to limited improvement in model recognition accuracy.

YOLOv8, as the latest generation of target detection model released by Ultralytics, has the advantages of high detection speed, high accuracy and simple deployment[7]. This paper optimizes the YOLOv8 model through multiple strategies to enhance its ability to capture multi-scale pest features and adapt to complex backgrounds. A diverse dataset containing ten common crop pests is constructed to verify the model's performance, providing a practical technical solution for the automatic monitoring of agricultural pests.

II. Related Methods And Model Improvement

A. YOLOv8 Network Structure

YOLOv8 is a multi-task model supporting object detection, instance segmentation and image classification, which optimizes the network structure on the basis of YOLOv5[8]. The model architecture is composed of three core parts: Backbone, Neck and Head. The Backbone uses C2f module instead of the traditional C3 module, which adopts a multi-branch flow design to provide richer gradient information and enhance feature extraction ability. The Neck adopts the PAN-FPN (Path Aggregation Network - Feature Pyramid Network) structure, which realizes multi-scale feature fusion through top-down and bottom-up bidirectional path aggregation, enhancing the expression ability of shallow detail features and deep semantic features. The Head uses an anchor-free design and a decoupled head structure, separating the classification and boundary box regression tasks into different branches, which improves the model's ability to handle complex targets[9].

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B. Model Improvement Strategy

To address the limitations of the original YOLOv8 model in multi-scale feature extraction and key feature focusing, three optimization strategies are proposed, forming a comprehensive improvement mechanism of "input enhancement-feature refinement-channel optimization" to improve pest recognition performance:

- Modifying Input Size: The input size of the original YOLOv8 model is 320×320 pixels. In this study, it is adjusted to 640×640 pixels. Larger input size can retain more image detail information, especially for small-sized pests and pests with inconspicuous morphological characteristics, which helps the model to accurately capture subtle features such as pest textures and edges, and improves the positioning accuracy of the bounding box. Although this adjustment increases the computational complexity to a certain extent, it significantly enhances the model's recognition ability for complex samples.

- Adding SE Attention Mechanism: SE (Squeeze-and-Excitation) attention mechanism is a lightweight channel attention module[10], which is designed to adaptively adjust the weight of each feature channel by learning the importance of feature channels. In this study, four layers of SE attention mechanism are added to the Head part of the original YOLOv8 model. The SE module first compresses the spatial dimension of the feature map through global average pooling to obtain the channel descriptor; then uses a multi-layer perceptron to model the dependence between channels and generate channel weight coefficients; finally, weights the original feature map through the Sigmoid activation function to enhance the model's attention to pest-related feature channels and suppress redundant background information. This improvement effectively enhances the model's ability to integrate deep semantic features in the classification and regression stages.The structural comparison of the detection head before and after the introduction of the SE attention mechanism is shown in Figure 1.

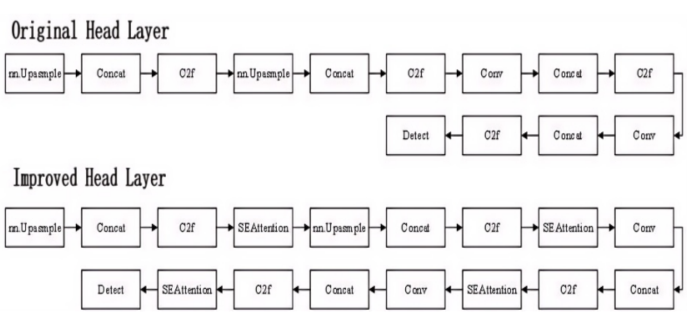

Fig. 1. Head Structure Comparison Before and After Adding SE Attention Mechanism

- Adding ECA Attention Mechanism: ECA (Efficient Channel Attention) attention mechanism is an efficient channel attention module that optimizes the calculation method of channel dependence[11]. Different from the SE mechanism that uses fully connected layers, the ECA module uses 1D convolution with adaptive kernel size to model the local cross-channel interaction, which not only reduces the computational cost but also maintains the performance. A layer of ECA attention mechanism is added to the Backbone part of the original YOLOv8 model, specifically after the C2f module of the shallow network. This improvement enhances the model's capture ability for small target features in the early feature extraction stage. The ECA module sets the output channel to 128, which is consistent with the number of channels of the previous C2f module output, ensuring seamless integration. By adaptively adjusting the channel weight, the model can focus more on the effective feature channels related to pests in the shallow feature map, improving the efficiency of feature selection.The structural comparison of the detection backbone before and after the introduction of the ECA attention mechanism is shown in Figure 2.

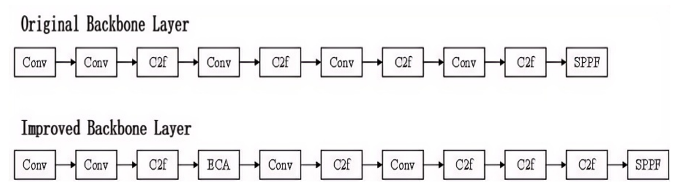

Fig. 2. Backbone Structure Comparison Before and After Adding ECA Attention Mechanism

C. Dataset Construction

- Dataset Source and Composition: A multi-source crop pest dataset is constructed by collecting images from the Kaggle platform. The dataset contains 2,470 images of ten common crop pests that seriously threaten food security in China, including mole cricket, sugar beet weevil, fall armyworm, Bactrocera dorsalis, rice leafhopper, Naranga aenescens, Serica orientalis, Lygus pratensis, Haplothrips tritici and Tryporyza incertulas. These images cover different shooting angles, lighting conditions and pest growth stages, ensuring the diversity and representativeness of the data. The dataset is divided into a training set, a validation set and a test set according to the ratio of 70:15:15. The training set contains 1,730 images for model parameter learning; the validation set contains 370 images for adjusting model hyperparameters and monitoring overfitting; the test set contains 370 images for final performance evaluation. After labeling, a total of 2,500 bounding boxes are obtained, and the specific composition of the dataset is shown in Table 1.

TABLE I. COMPOSITION OF CROP PEST DATASET

Dataset	**Number of Images**	**Number of Bounding Boxes**
Training set	1730	1760
Validation set	370	370
Test set	370	370
Total	2470	2500

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- Data Preprocessing: To enhance the generalization ability and robustness of the model, a series of preprocessing operations are implemented on the dataset: first, all images are uniformly resized to 640×640 pixels to meet the input requirement of the improved model; second, data augmentation techniques such as random horizontal flipping (probability 0.5), random rotation (0-15°) and brightness adjustment (0.8-1.2 times) are applied during the training phase to expand the dataset volume and alleviate overfitting; third, the pixel values of all images are normalized to the range of [0, 1] to accelerate model convergence. The LabelImg tool is used for data annotation, and the annotation results are saved in txt format, including target category, center coordinates of bounding box, width and height.

III. Experimental Setup And Results

A. Experimental Environment and Training Parameters

The experiment is conducted on a PC equipped with Microsoft Windows 11 operating system, an AMD Ryzen 77735H CPU and an AMD Radeon Graphics GPU. The software environment is configured as Python 3.8, PyTorch deep learning framework, OpenCV-Python image processing library and NumPy numerical calculation library. The training parameters are set as follows: input image size 640×640×3, training epochs 120, batch size 16, optimizer SGD (Stochastic Gradient Descent), initial learning rate 0.01, and data loading thread 0. Transfer learning is adopted to accelerate model convergence, and the pre-trained weight of YOLOv8 is used as the initial weight.

B. Performance Evaluation Metrics

A multi-metric evaluation framework is employed to comprehensively assess the model's performance, including:

- Precision (P), defined as the ratio of correctly identified positive samples to all samples predicted as positive, reflecting the accuracy of positive class predictions;

- Recall (R), representing the proportion of correctly detected positive samples out of all actual positive samples, evaluating the completeness of positive class identification;

- MAP (mean Average Precision), the average of AP values of all categories, comprehensively reflecting the model's overall detection performance;

- Loss values, including training loss (train/loss) and validation loss (val/loss), quantifying the discrepancy between the model's predicted outputs and true values.

C. Experimental Results and Analysis

- Model Performance Comparison: Four models are compared under the same experimental conditions, and the results are shown in Table 2. The original YOLOv8 model achieves a precision of 80%. After modifying the input size to 640×640, the precision increases by 2% to 82%, indicating that larger input size can effectively retain detail features and improve recognition accuracy. Adding the ECA attention mechanism to the Backbone improves the precision to 83.5%, verifying that the ECA module can enhance the model's feature extraction ability in the shallow network. The model integrated with the SE attention mechanism achieves the highest precision of 92%, which is 12% higher than the original model, and its recall and mAP also show significant improvements, demonstrating that the SE module can effectively enhance the model's attention to key feature channels. Although the inference time of the improved models is slightly longer than that of the original model, it still meets the real-time requirements of practical applications.

TABLE II. PERFORMANCE COMPARISON OF DIFFERENT MODELS

Model	Precision (%)	Recall (%)	mAP50 (%)	Inference Time (s/Image)
YOLOv8	80.0	78.5	79.2	0.21
YOLOv8-640	82.0	80.3	81.5	0.22
YOLOv8-ECA	83.5	81.7	82.8	0.23
YOLOv8-SE	92.0	89.5	90.7	0.25

- Loss Curve Analysis: The loss curves of the models are shown in Fig. 3. With the increase of training epochs, the training loss and validation loss of all four models exhibit a downward trend, indicating that the models are constantly learning and converging. The loss curve of the original YOLOv8 model shows obvious fluctuations in the later training stage, and there is a certain gap between the training loss and validation loss, suggesting mild overfitting. In contrast, the loss curves of the improved models are more stable. Especially the YOLOv8-SE model has the smoothest loss curve, and its training loss and validation loss reach the lowest values and tend to stabilize in the later stage, proving that the integration of the SE attention mechanism can effectively enhance the model's feature learning ability and avoid overfitting.

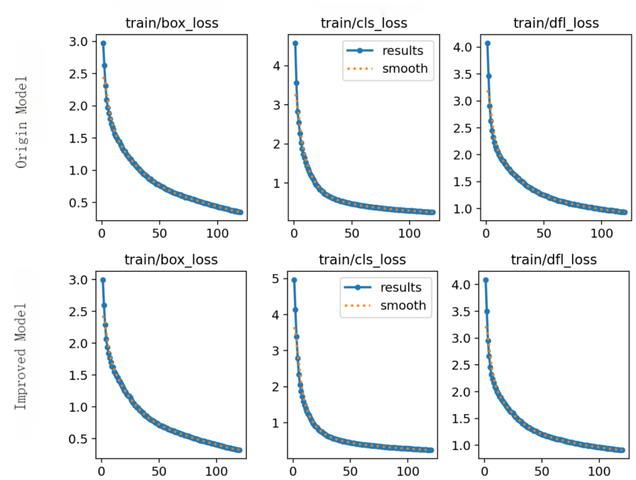

Fig. 3. Loss Curves of Different Models

- Classification Performance of Each Pest Category: To further analyze the performance of the improved model, the precision of the YOLOv8-SE model for each pest category in the test set is calculated. The results show that the model achieves high recognition accuracy for all ten

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pest categories: the precision for beet weevil and rice leafhopper reaches 91%, the precision for mole cricket is 88%, and the precision for other categories is above 85%. The average precision of all categories is 89.7%, demonstrating that the model is capable of accurately distinguishing between different pest categories. Misclassifications are rare and mainly occur between pests with similar morphological characteristics, but the extremely low misclassification rate has no negative impact on practical application.

- Robustness Analysis: The YOLOv8-SE model performs well on different types of samples. For pests with small sizes (such as Haplothrips tritici), the average recognition accuracy is above 85%; for pests with obvious morphological characteristics (such as sugar beet weevil), the accuracy reaches more than 90%. Even under interference from complex backgrounds or image blurring, the model can maintain high confidence, demonstrating strong robustness and adaptability to practical application scenarios.

IV. CONCLUSION AND OUTLOOK

This paper proposes an improved crop pest recognition method based on YOLOv8, which optimizes the model through three strategies: modifying input size, adding SE attention mechanism and adding ECA attention mechanism. A diverse dataset containing ten common crop pests is constructed to verify the model's performance. Experimental results show that the improved model achieves significant performance improvement, with the YOLOv8-SE model reaching the highest precision of 92%, which is 12% higher than the original model. The method has fast inference speed and strong robustness, meeting the needs of rapid and accurate identification of crop pests.

This study has certain limitations: the dataset covers only ten common crop pests, and the model is trained and tested on a PC. Future work will focus on three aspects: first, expanding the dataset to include more pest types and samples under different environmental conditions to improve the model's generalization ability; second, optimizing the model's lightweight design to reduce computational complexity and realize deployment on edge devices such as drones and embedded systems; third, integrating pest occurrence rules and prevention suggestions to provide comprehensive technical support for agricultural production management.

ACKNOWLEDGMENT

This work was supported by the Nanyang Science and Technology Key Research Project (Grant No.: 24KJGG048), the 2025 Educational Teaching Reform Research and Practice Projects of Nanyang Institute of Technology (General Funded Projects), including Grant No. NIT2025JY-017 entitled "Research and Practice of Closed-Loop Teaching of 'Goal-Design-Evaluation' for the Smart Course of Sensor and Detection Technology under the OBE Concept" and Grant No. NIT2025JY-085 entitled "Exploration and Practice of Teaching Methods for 'Principle and Application of Single-Chip Microcomputer' Empowered by Artificial Intelligence"; and the 2025 Smart Course Construction Project of Nanyang Institute of Technology for the course "Sensor and Detection Technology".

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